Analysis Choices and Ranking Stability in Kinase Inhibitor Selectivity Scores

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Abstract

Kinase inhibitor selectivity is summarized by several competing scores, including the S-score, selectivity entropy, Gini coefficient and ratio-based measures, which are often used interchangeably. We asked which analysis choices change the resulting compound rankings, on four profiling datasets spanning three assay technologies, each curated to one reference target per human protein kinase. Agreement between the scores depends on how affinities are weighted, which compounds are ranked, and how pairs without detected binding are treated. With entropy and Gini on a common log-affinity scale, the S-score, entropy and Gini agree more with one another than with the ratio among compounds with an active kinase. The entropy as originally published, with association-constant weights, does not. Moving the activity boundary between the floor and the activity threshold reverses 8–13% of compound pairs under the S-score, entropy and Gini on three datasets, and over 20% on the fourth, whose rankings largely follow how many kinases were measured. We state three criteria for a selectivity measure: activity eligibility, consistency across placements of the activity boundary, and monotonicity when a weakly bound kinase is added. No boundary parameter we tested meets the second, so the boundary must be fixed in the analysis protocol and its effect reported.

Monotonicity holds for simple measures such as a count of active kinases and for a gated, smooth-hinge entropy we construct, which on these data ranks compounds almost as the share of targets with detected binding does. On random subpanels, scores that rank compounds by this share recover rankings from fewer targets than the ratio and the published entropy, but some compounds can no longer be scored.

Keywords: binding selectivity, cancer drug discovery, cheminformatics, kinase inhibition, research methodology.

Introduction

Kinase inhibitors are among the most intensely investigated compound classes in drug discovery, with 94 FDA-approved agents and several hundred in clinical development.[1] The human kinome comprises more than 500 proteins with a conserved ATP-binding site,[2] and selectivity is a primary criterion in lead optimization.[3] Profiling assays measure a compound against a panel of tens to hundreds of kinases, and the resulting profile is often reduced to one selectivity score. Several scores are in use: the S-score, the fraction of the panel bound below an affinity cutoff;[4] the selectivity entropy, the Shannon entropy of the profile treated as a distribution;[5] the Gini coefficient of the profile;[6] the partition index, the share of binding that falls on one target;[7] and ratio-based measures, which compare the primary target with an off-target. They are often used interchangeably, although each summarizes a different aspect of the profile.

Earlier work established that these scores can disagree. Bosc et al.[8] compared six scores on the Davis and Anastassiadis datasets, computed several at more than one parameter value, compared them on compounds and kinases shared between datasets, and found that compound rankings differ substantially between them. They traced individual disagreements to profile shape, such as a broad binder with one dominant target. The KInhibition platform[9] noted that conflicting results across metrics are common and introduced a composite score,

and a later resource from the same group offers a choice of scores.[10] Miljkovic and Bajorath[11] found the concentration-dependent selectivity score of Klaeger et al.[12] broadly consistent with activity data from ChEMBL. Karaman et al.[4] and Uitdehaag et al.[13] showed that the S-score and the entropy of individual compounds vary when they are computed on random subpanels.

What has not been characterized systematically is how analysis choices alter agreement between the scores: how affinities are weighted, which compounds are ranked, how pairs without detected binding are treated, which assay targets count as kinases, and where each score's parameter is set. It is also not known which associations between profile shape and ranking instability persist across these choices, or how much of a ranking a smaller panel retains once some compounds can no longer be scored. The study concerns scores computed from a binding profile alone.

This study adds a systematic account of these choices. We compare the S-score, entropy, Gini and ratio on four datasets that span three assay technologies, each curated to one reference target per human protein kinase, with the entropy and Gini both in a common log-affinity form and in their published forms. We vary the compounds ranked, the treatment of pairs without detected binding and each score's parameter, and relate the resulting instability to features of the binding profile. From these results we state three criteria for a selectivity measure: activity eligibility, consistency across placements of the activity boundary, and monotonicity when a weakly bound kinase is added. We evaluate the existing scores and simple comparators against them, and give a gated, smooth-hinge entropy with a proof of the third. Finally, we measure how well each score's ranking is recovered from random subpanels, and for how many compounds. Stating such criteria follows the axiomatic approach Shannon used for entropy.[14]

Methods

Datasets, Target Selection and Missing Values

Table 1 summarizes the four datasets. The Davis dataset[15] reports dissociation constants from a competition-binding assay, and the Klaeger dataset[12] apparent dissociation constants from Kinobeads chemoproteomics in cell lysate. Both were converted to $\mathrm{p}K_d = -\log_{10}(K_d/10^9)$.[16] The Anastassiadis dataset[17] reports residual kinase activity in a radiometric assay at 0.5 μM inhibitor, which we converted to percent inhibition (100 minus residual activity, clipped to $[0, 100]$). The Metz dataset[18] reports $\mathrm{p}K_i$ values from a medicinal-chemistry campaign. Each score is computed on the native readout of its dataset, and datasets are compared only through rank correlations computed within each.

Table 1: The four datasets. The floor x_0 is the value given to pairs with no detected binding (Davis, Klaeger) or not tested (Metz); on Anastassiadis it is an operational cutoff for weak inhibition. τ^* is the activity threshold.

Dataset	Assay; readout	Compounds	Targets (released, curated)	x_0	τ^*
Davis	competition binding; $\mathrm{p}K_d$	68	433, 363	5.0	6
Klaeger	Kinobeads in lysate; $\mathrm{p}K_d$	222	343, 217	5.0	6
Anastassiadis	radiometric activity; % inhibition	178	300, 291	20%	50%
Metz	biochemical activity; $\mathrm{p}K_i$	704	172, 172	4.0	6

Target selection. Selectivity across kinases needs one entry per kinase, chosen without reference to any compound's affinities. The released Davis panel lists 433 constructs. They include mutant, phosphorylation-state and single-domain constructs of the same gene (12 for EGFR), 20 lipid kinases and three non-human kinases, and for 26 of the 68 compounds the two best-bound constructs are forms of one gene. We keep one reference construct per human protein-kinase gene: the wild-type construct, or its catalytic or N-terminal kinase domain where several domains are profiled. Genes profiled only as mutants are removed. The 343 Klaeger targets are the proteins the Kinobeads pulldown quantified, including complex subunits, such as cyclins, captured with their kinases. We keep the 217 single-gene

protein kinases that Klaeger et al. annotate, except Q6ZSR9, which UniProt lists as an uncharacterized protein without a kinase domain. This removes lipid kinases, four protein groups, and BCR and TRRAP, which the source does not annotate as kinases and flags as possibly co-enriched with ABL1 and CDK8. On Anastassiadis we keep the first-listed of nine kinases profiled in two complexes or isoforms. The Metz columns are already one human protein kinase each. The code repository lists every target with the reason it was kept or removed, and Supporting Information (Section S1) compares the released panels.

Floor-coded and missing values. Several kinds of pair receive the same floor value x_0, which is not a measured detection limit. On Davis, the released matrix assigns $\mathrm{p}K_d = 5.0$ ($K_d = 10$ μM) to pairs with no detected binding, following DeepDTA;[16] Bosc et al. used nearly the same value (10.1 μM).[8] On Klaeger, we kept the 5,123 drug–target pairs that the original authors classified as high confidence. We coded at $\mathrm{p}K_d = 5.0$ three kinds of pair: those with no detected binding, those whose fit was classified as low confidence, and the 194 measured affinities weaker than 10 μM. A low-confidence fit does not establish weak binding: of the 405 such pairs on curated targets, 375 have a fitted $\mathrm{p}K_d$ above 5. Giving all of these pairs the floor value is an analysis convention. On Metz, pairs that were not tested are coded at $\mathrm{p}K_i = 4.0$, the lowest value reported in the data. A median Metz compound is measured against only 21 of the 172 kinases, so we kept the 704 compounds measured against at least 50 (median 64). On Anastassiadis, the 1.1% of missing entries were set to 0% inhibition. There we place x_0 at 20% as an operational cutoff for weak inhibition, since 54% of the entries lie between 0% and 20%. A compound is active against a target above the activity threshold τ^*.

Selectivity Scores and Their Implementations

A compound's profile is the vector $\mathbf{x} = (x_1, \ldots, x_N)$ of its readouts across the N panel targets, with higher values for tighter binding or stronger inhibition. Every score is oriented so that

higher values are more selective (Table 2).

The S-score of Karaman et al.[4] is the fraction of the panel bound above a threshold,

$$S(\tau) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[x_i > \tau], \tag{1}$$

negated. They reported K_d cutoffs of 3 μM and 100 nM, which lie within our sweep of τ. The selectivity entropy of Uitdehaag and Zaman[5] is the Shannon entropy $H = -\sum_i p_i \log_2 p_i$ of the profile normalized to a distribution, negated. In its published form, $p_i = K_{a,i} / \sum_j K_{a,j}$, with association constants $K_{a,i} = 1/K_{d,i}$. To place all four definitions on one scale, we also use a log-affinity form (on Anastassiadis, percent inhibition), which weights each target by its readout above a baseline β:

$$p_i = \frac{\max(x_i - \beta,\, 0)}{\sum_j \max(x_j - \beta,\, 0)}. \tag{2}$$

The baseline is a parameter we introduce; it has no published reference values. Graczyk's Gini coefficient[6] measures how unequally binding is spread over the panel. Its published form uses percent-inhibition values. We compute it on the same baseline-subtracted weights,

$$G(\beta) = \frac{2 \sum_{i=1}^{N} i \cdot w_{(i)}}{N \sum_{i=1}^{N} w_{(i)}} - \frac{N+1}{N}, \qquad w_{(i)} = \max(x_i - \beta,\, 0) \text{ sorted ascending}, \tag{3}$$

and on Anastassiadis also in its native form. The ratio compares the primary target with the k-th ranked off-target,

$$R_b(k) = x_{(1)} - \max(x_{(k+1)}, b), \tag{4}$$

where $x_{(j)}$ is the j-th largest value. The level b is the lowest coded value (5.0 on Davis and Klaeger, 4.0 on Metz and 0% on Anastassiadis), except in the recovery analysis, where it is x_0, and in D2, where it is swept. At this level $R_b(k)$ equals Bosc et al.'s ranking score at rank $k+1$.[8] We cap k at 5 because half of the Klaeger compounds have fewer than five active kinases. Unless stated otherwise, the entropy and Gini are in their log-affinity forms.

Table 2: Score implementations. Parameter ranges are for the pK_d datasets; on Anastassiadis τ runs over $[50, 90]\%$ and β over $[0, 30]\%$, and on Metz over $[5.0, 7.5]$ and $[4.0, 5.5]$. The last four rows are comparators.

Score	Weights or formula	Parameter swept	No target above the parameter
S-score	fraction above τ	$\tau \in [5.5, 8.0]$	score 0
Entropy, log-affinity	$\max(x_i - \beta, 0)$	$\beta \in [5.0, 6.5]$	undefined; set to 0 (top rank)
Gini, log-affinity	$\max(x_i - \beta, 0)$	$\beta \in [5.0, 6.5]$	undefined; set to 0 (bottom rank)
Ratio $R_b(k)$	$x_{(1)} - \max(x_{(k+1)}, b)$	$k = 1, \ldots, 5$	uses b
Entropy, published	$K_{a,i}$	none	—
Gated entropy	gated softplus (below)	none	not scored (D1)
Partition index $P_{\max}$	$\max_i K_{a,i} / \sum_j K_{a,j}$	none	—
Active count	number above τ^*	none	not scored (D1)
Entropy, $\beta = \tau^*$	$\max(x_i - \tau^*, 0)$	none	not scored (D1)
S-score, $\tau = x_0$	fraction above x_0	none	not scored (D1)

Conventions. Where the entropy and Gini are undefined, we set both to 0. This ranks the compound as maximally selective under the entropy and minimally selective under the Gini. $R_b(k)$ uses b for a missing off-target and so scores every compound, as Bosc et al.'s ranking score does. Log-affinity weights with $\beta = x_0$ give floor-coded pairs no weight. Association constants give them a small positive weight if they are kept at the floor value. For the published entropy and $P_{\max}$ we give them no weight. As sensitivity analyses we keep them at the floor value and, on Klaeger, use every measured high-confidence affinity at its measured value, including those weaker than the floor; neither of the first two treatments is free of assumptions.

Comparators. The active count, the entropy with $\beta = \tau^*$ and the S-score with $\tau = x_0$ (the share of the panel with detected binding) are scored only for compounds with a target above τ^*. The partition index of Cheng et al.[7] is taken at the most strongly bound target, as Uitdehaag and Zaman did.[5] The CATDS score of Klaeger et al. needs a designated target and Kinobeads dose–response curves, which the other three datasets lack, so it is not compared across datasets; a proxy is compared in Supporting Information (Section S6).

Ranking Comparisons and Sensitivity Analysis

For each score and parameter value, compounds are ranked from most selective (rank 1), with ties given their average rank. Ranks are always computed among a stated set of compounds. Analyses of compounds with an active kinase rank them among themselves, which applies D1 (below); analyses that compare compounds with and without an active kinase rank all compounds. Agreement between two definitions is the Spearman correlation between their median rank vectors over each definition's parameter sweep: τ in steps of 0.25 (11 values) and β in steps of 0.25 (7 values) on the pK datasets, τ in steps of 5% (9 values) and β in steps of 5% (7 values) on Anastassiadis, and $k = 1, \ldots, 5$. Using mean ranks instead changes no all-compound correlation by more than 0.08.

A compound's rank instability is the standard deviation of its rank across a sweep, within one definition or pooled over all four definitions' sweeps. Pooling weights each definition by its number of parameter values, so the S-score carries the most weight. Profile features (the number of active kinases, the top_1–top_2 affinity gap, the second-ranked affinity, and the standard deviation, range and skewness of the active affinities) are correlated with within-definition instability by Spearman correlation. A feature undefined for a compound with too few active kinases is set to 0, and a check excludes these compounds. These associations are exploratory: a Bonferroni correction covers only the 24 tests per dataset of the feature table, and p-values are otherwise uncorrected. Bootstrap intervals use 1,000 resamples of compounds.

To compare the effect of the dataset with that of the definition, we matched 46 compounds profiled in both Davis and Klaeger by PubChem CID, restricted both panels to the 201 kinases they share (matched by gene symbol, with renamed symbols mapped to current ones), and correlated each definition's rankings between the datasets and the definitions' rankings within each dataset. Analyses used Python 3 with NumPy, pandas and SciPy;[19–21] Supporting Information (Table S9) names the script behind every result.

Evaluation Criteria and the Gated Entropy

A selectivity measure is a function f of a profile, defined for every panel size N, with higher values more selective. Let $B = [x_0, \tau^*]$, the interval between the floor and the activity threshold, where an analyst must decide how much weak binding counts.

D1 (activity eligibility). $f(\mathbf{x})$ is undefined, or flagged as not eligible, whenever $\max_i x_i \leq \tau^*$. A compound with no target above τ^* has no target against which to judge selectivity. D1 is a reporting rule and is assessed from each score's reporting convention. Bosc et al. withheld ranks for such compounds under the S-score only;[8] D1 applies the rule to every score.

D2 (distributional consistency). Let θ be the parameter that places a score's activity boundary: the S-score's τ, the entropy's and Gini's β, the gated entropy's gate x_0 (with β moved with it and T fixed), and the ratio's level b. A pair of compounds *reverses in* B if $f(\mathbf{x};\theta) - f(\mathbf{y};\theta)$ is positive at some $\theta \in B$ and negative at another. D2 requires that at most $\varepsilon = 1\%$ of the pairs of compounds scored under D1 reverse in B, on every dataset and, for the ratio, for every k. A score that fixes such a parameter, such as the active count, is assessed with the parameter swept, so D2 concerns a family of scores. D2 is a boundary-sensitivity diagnostic. Whether a score should satisfy it depends on the use: a score whose question changes with the threshold can reasonably give different rankings at different thresholds. Fixing the boundary defines the analysis protocol; it does not remove the sensitivity, which D2 reports. The S-score is evaluated at every observed value in B, where its value changes. The gated entropy's gate is evaluated there and just below each observed value, where its score jumps, and on 201 equally spaced values; the other parameters are evaluated on the 201 values. Apart from the S-score, the percentages are therefore estimates on a grid.

D3 (monotonicity under weak-target addition). For any $\mathbf{x}$ with $\max_i x_i > \tau^*$, if $\mathbf{x}' = (x_1, \ldots, x_N, x_{N+1})$ with $x_0 < x_{N+1} \leq \tau^*$, then $f(\mathbf{x}') \leq f(\mathbf{x})$. Otherwise a compound can be made to look more selective by profiling it against more kinases it binds weakly. D3 is assessed from each score's functional form. It concerns adding a target, not strengthening

one already in the profile.

We state no criterion for the ratio's choice of reference off-target, because different references answer different questions.

The gated entropy. For a profile with $\max_i x_i > \tau^*$ (undefined otherwise, as D1 requires), define

$$w_i = \mathbf{1}[x_i > x_0]\; T \log\bigl(1 + e^{(x_i - \beta)/T}\bigr), \qquad p_i = \frac{w_i}{\sum_j w_j}, \qquad f(\mathbf{x}) = \sum_i p_i \log_2 p_i, \tag{5}$$

with the emphasis baseline $\beta = x_0$ and the smoothing width $T = \tau^* - x_0$ (1 pK unit on Davis and Klaeger, 2 on Metz, and 30 percentage points on Anastassiadis). It is the negative Shannon entropy of a gated, smoothed profile. The non-binder gate $\mathbf{1}[x_i > x_0]$ gives floor-coded pairs no weight. The softplus replaces the entropy's hard cutoff $\max(x_i - \beta, 0)$, which gives a kinase just above β a weight close to zero.

Proposition. With $\beta = x_0$ and $T \geq (\tau^* - x_0)/\ln(2^e - 1) \approx 0.58\,(\tau^* - x_0)$, adding a kinase with a value in $(x_0, \tau^*]$ never raises f. With $T = \tau^* - x_0$ this holds for every $\beta \in B$.

The proof (Supporting Information, Section S4) uses a lower bound on the weights. Every gated target has weight above $m = T \ln 2$, and the added target has weight at most $e\,m$. This bound on its share of the total weight is enough for the entropy not to decrease. The bound on T cannot be lowered. The same minimum weight makes the score discontinuous at x_0: for a profile with one target at p$K_d = 7$, a second target at the floor leaves the score at 0, and one just above the floor lowers it by 0.80 bits. To measure the effect of noise near the gate, we perturbed the measured values above the floor (normal noise with standard deviation 0.1 and 0.3 pK units), set values pushed below the floor to the floor, and left floor-coded pairs fixed. On Anastassiadis every entry was perturbed, with standard deviation 3 and 9 percentage points. Each setting used 50 draws, and D1 was applied again to the noisy profiles.

Recovery from Smaller Panels

Subpanels of m targets were drawn uniformly at random from the curated targets, 50 per size, at $m = 10$, 20, 30, 40, 50, 65, 80, 100, 125, 150, 200, 250 and 300 below the full panel, and every score was computed on the same subpanels. D1 was applied to every score on every subpanel. Coverage is the share of compounds scored on the full panel that are still scored on the subpanel; it is the same for every score. Recoverability is the share of clearly separated full-panel pairs (ranks at least a quarter of the scored compounds apart) that are scored on the subpanel and keep their order, with a pair tied on the subpanel counted as one half. Conditional agreement is the Spearman correlation between subpanel and full-panel scores over the compounds scored on both. Each score is compared with its own full-panel ranking, and its clearly separated pairs are chosen from that ranking, so recovery measures how much of a score's own ordering a subpanel keeps, not which score is more accurate. We write p^* for the smallest tested size at which the mean conditional agreement over subpanels exceeds 0.90, a descriptive benchmark that we chose. As a diagnostic, each compound's rank error at 50 targets is correlated with the largest change in its full-panel score when one target is left out. Results without the D1 gate, which mix ranking with the conventions for undefined scores, are in Supporting Information (Section S5).

Results

Analysis Choices Change Agreement Between Scores

With the entropy and Gini in their log-affinity forms, and compounds with an active kinase ranked among themselves, the S-score, entropy and Gini agree more with one another than with the ratio (Table 3). On Davis, where every compound has an active kinase, they correlate at 0.963–0.999, and the ratio with each at 0.51–0.56. Among the 200 Klaeger compounds with an active kinase, they correlate at 0.89–0.997, and the ratio with each at 0.38–0.64. On

this scale the S-score, entropy and Gini all summarize how activity is spread over the panel, while the ratio compares the primary target with one off-target.

Table 3: Spearman correlations between median rankings. S: S-score; E: entropy; G: Gini; R: ratio, with the entropy and Gini in their log-affinity forms. E_{K_a}: the published entropy, with floor-coded pairs given no weight and, in parentheses, kept at the floor. In the active columns, compounds are ranked among themselves. Every Davis compound has an active kinase.

	Davis	Klaeger	
Pair	all ($n = 68$)	all ($n = 222$)	active ($n = 200$)
S–E	0.963	0.896	0.897
S–G	0.963	0.508	0.892
E–G	0.999	0.658	0.997
S–R	0.563	0.078	0.378
E–R	0.513	0.344	0.622
G–R	0.510	0.655	0.635
E_{K_a}–S	0.738 (0.344)	0.571 (−0.597)	0.564 (−0.412)
E_{K_a}–G	0.736 (0.368)	0.530 (0.077)	0.782 (−0.094)
E_{K_a}–R	0.861 (0.688)	0.657 (0.648)	0.916 (0.591)

Weighting. The published entropy, with association-constant weights, correlates more with the ratio than with the S-score or Gini on Davis, Klaeger and Metz, under both treatments of floor-coded pairs (Table 3; Supporting Information, Table S2). How far it is from the S-score and Gini depends on that treatment. Floor-coded pairs make up 71% of the Davis matrix, and the published entropy's correlation with the S-score there is 0.34 with them at the floor and 0.74 with them given no weight. The published and log-affinity entropies correlate at 0.37 (Davis) and −0.31 (Klaeger) with floor-coded pairs at the floor, and at 0.74 and 0.79 with them given no weight (Figure 1); the negative Klaeger value comes from the floor-coded pairs. Using every measured Klaeger affinity at its measured value, including the 194 weaker than the floor, changes the published entropy's ranking little (Spearman 0.9996 with the floor-excluded ranking among active compounds) and changes its correlations and recovery by at most 0.02 (Supporting Information, Section S2). On Anastassiadis, the native percent-inhibition Gini correlates with the entropy at 0.63, and with the S-score and the

ratio at only 0.34 and 0.33. The grouping of the S-score, entropy and Gini is therefore a property of the log-affinity form.

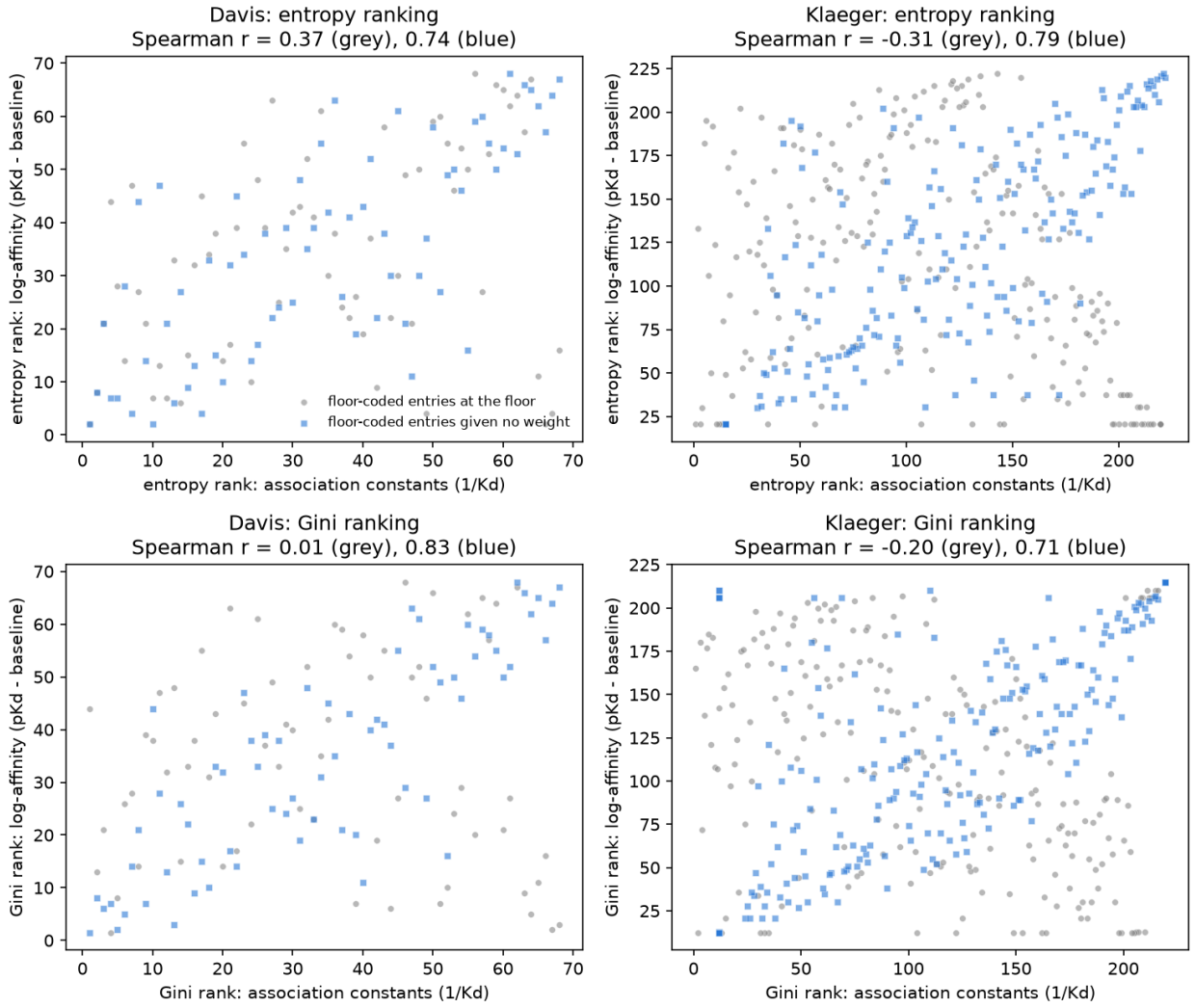


Figure 1: Rankings under association-constant weights (x-axis) against the baseline-subtracted log-affinity form (y-axis; median rank over the baseline sweep), on Davis and Klaeger. Top: entropy, for which association-constant weights are the published form. Bottom: Gini, shown for comparison; its published form uses percent inhibition. Grey: floor-coded pairs kept at the floor; blue: floor-coded pairs given no weight. Spearman r above each panel.

Compounds ranked. Twenty-two Klaeger compounds have no target above the activity threshold. With them included, the Gini correlates more with the ratio (0.66) than with the S-score (0.51), although the difference is not significant (bootstrap 95% interval of the difference $[-0.35, 0.06]$, against $[0.16, 0.36]$ among active compounds). Each definition scores

these compounds by its own convention, and the definitions disagree about them. The S-score and entropy rank them among the most selective (median ranks 24 and 20 of 222) and the ratio near the bottom (182). The Gini ranks them near the top where its score is defined and near the bottom where it is not (mean ranks 32 and 212). Pooled across the four definitions' sweeps, their rank instability is $\sigma = 77.9$, against 32.2 for active compounds ($p < 0.001$, Mann–Whitney U test). This contrast comes from the definitions disagreeing. Within a single definition it is significantly higher only for the Gini (65.0 against 16.3), and it reverses for the S-score (19.0 against 25.7, $p = 0.01$), whose score is 0 for all of these compounds at thresholds of 6 or above. Sixteen of the 22 bind some kinase above the floor. Computed only over the configurations where their scores are defined, these 16 have lower mean instability than the active compounds within the entropy and the Gini (7.0 against 13.5, and 6.7 against 13.6; both $p = 0.002$), although not every one of them does. The other six show no binding to any curated target, although all are established inhibitors; for three of them the target was quantified for other compounds (Supporting Information, Section S3). Whether a compound is eligible for a selectivity score therefore depends on the panel and the threshold. Supporting Information (Section S3) gives the per-definition contrasts.

Other datasets. Among compounds with an active kinase, the S-score, entropy and Gini correlate at 0.81–0.999 on all four datasets, and the ratio less with each (Supporting Information, Figure S1). On Anastassiadis the separation is weak: the ratio correlates with the other three at 0.79–0.84, against 0.87–0.999 among them. With all compounds included, this falls to 0.48–0.55, because of the 45 compounds with no active kinase. On Metz the ratio correlates with the other three at only 0.15–0.20, and this reflects missing measurements. Nearly every measured value (99.9%) lies above the floor, so a compound measured against more kinases has more kinases with positive weight and looks less selective. The Spearman correlation between a compound's coverage and its median rank (rank 1 = most selective) is 0.98 for the entropy, 0.95 for the Gini, 0.76 for the S-score and 0.11 for the ratio. Compounds

measured against more kinases are also active against a larger share of those measured, but that association is much weaker ($r = 0.33$). Metz therefore shows how floor imputation changes rankings, not how the definitions relate on complete profiles.

Target selection and dataset. On the released Davis panel, the ratio agrees less with the other definitions than on the curated panel (0.34–0.46 against 0.51–0.56; Supporting Information, Section S1). On the 46 compounds profiled in both Davis and Klaeger, the same definition's Davis and Klaeger rankings correlate, on the 201 kinases both panels share, at 0.34–0.84 (mean 0.61), and different definitions' rankings within one dataset at −0.08–1.00 (mean 0.57). Among the 40 compounds active in both datasets, the means are 0.72 and 0.73, and the Gini's cross-dataset correlation rises from 0.45 to 0.84. On these compounds the choice of definition moves rankings about as much as the change of dataset. The same compounds are active against 15.0% of the Davis panel and 3.9% of the Klaeger panel on average, and without restricting the panels to their shared kinases the cross-dataset correlations are lower (mean 0.53). Restoring the low-confidence Klaeger pairs shifts the all-compound Klaeger correlations by at most 0.08 and the active-compound correlations by at most 0.01, and leaves every p^* of the recovery analysis unchanged (Supporting Information, Section S1).

Parameter Sensitivity and Profile Associations

Moving each definition's activity boundary across B reorders a substantial share of compound pairs. The S-score's threshold reverses 8.4–13.0% of pairs of scored compounds on Davis, Klaeger and Anastassiadis, and 21.8% on Metz. The entropy's and Gini's baselines reverse 7.9–12.4%, and 22.5–22.8% on Metz. The ratio's level b, which replaces any off-target value below b, measured or not, reverses 2.1–13.7% on Klaeger and Anastassiadis depending on k, 0–2.6% on Davis and 0.04–1.03% on Metz. Sweeping the gated entropy's gate x_0, with $\beta = x_0$ and T fixed at $\tau^* - x_0$, reverses 9.2–14.4% of pairs, and 23.6% on Metz. On Klaeger

its per-compound instability matches that of the entropy's baseline (mean rank $\sigma = 6.8$ and 7.0; Supporting Information, Figure S2). With the gate fixed, its remaining parameter, the emphasis baseline, reverses at most 0.72% of pairs. Supporting Information (Table S3) gives every value.

Reference off-target. When an active compound's two best-bound kinases have near-identical affinities, the ratio depends heavily on k. $R(1)$ compares the two near-tied kinases and is close to zero, and $R(2)$, which skips past the tie, can be large. On the 22 Klaeger compounds with an active kinase and a top_1–top_2 gap below 0.05, mean $R(1) = 0.02$ against $R(2) = 0.50$. The two values answer different questions: $R(1)$ treats the second kinase as an off-target, and $R(2)$ treats it as a co-target. On Davis, fedratinib, with JAK2 and GAK tied, has the most unstable ratio ranking (Figure 2, left).

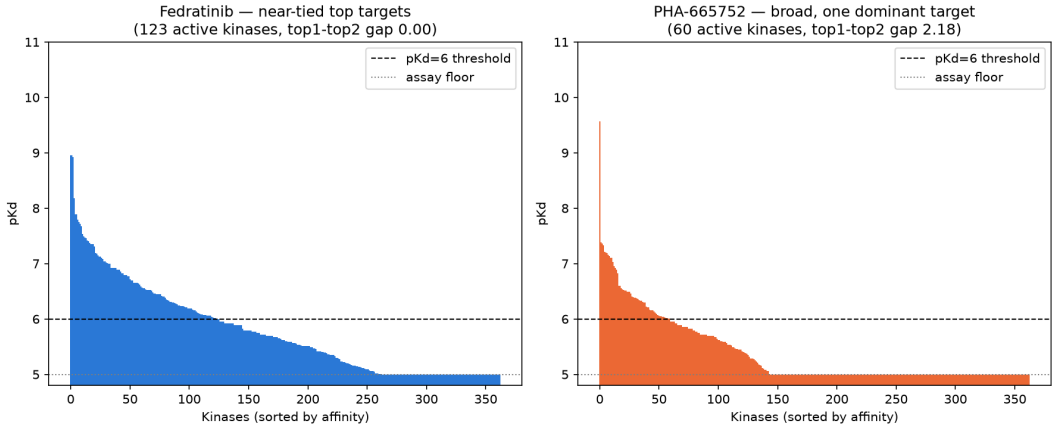


Figure 2: Binding profiles of two Davis compounds, targets sorted by affinity. Left: fedratinib, the most ratio-sensitive compound, with JAK2 and GAK tied among 123 active targets. Right: PHA-665752, the most unstable compound overall ($\sigma = 16.4$), with one dominant target among 60 active targets. Dashed line: $\text{p}K_d = 6$ activity threshold; dotted line: $\text{p}K_d = 5$ assay floor.

Profile associations. Ranked among compounds with an active kinase, three associations between profile features and instability hold on both $\text{p}K_d$ datasets (Supporting Information, Table S4). The number of active kinases correlates negatively with entropy instability on every dataset ($r = -0.30$ to -0.42; on Davis only nominally, $p = 0.01$) and with Gini instability on

Davis and Klaeger ($r = -0.31$ and -0.25). A possible reason is that moving the baseline changes the active set of such compounds. On Klaeger, entropy instability correlates with the share of a compound's above-floor kinases that lie within the baseline sweep ($r = 0.22$, $p = 0.002$). The top_1–top_2 gap correlates with ratio instability on Davis, Klaeger and Metz ($r = -0.44$, -0.28 and -0.26, all $p < 0.001$), but not on Anastassiadis ($r = 0.00$; -0.15, $p = 0.10$, without eight compounds whose two best targets are at the 100% inhibition ceiling). With a Bonferroni correction, only the ratio–gap association remains significant on both pK_d datasets. On Klaeger, S-score instability also correlates with the range of active affinities ($r = -0.32$; partial $r = -0.22$ with the number of active kinases held fixed), but not on Davis. Ranked among all 222 Klaeger compounds, the skewness of the active affinities was the strongest correlate of Gini instability ($r = +0.26$); ranked among active compounds it is not significant ($r = +0.07$, $p = 0.31$), so the marginal association came mainly from the scoring of compounds with no active kinase. With the number of active kinases held fixed, a weaker association remains (Supporting Information, Section S3).

Broad profiles. Among active compounds, seven of the ten largest entropy–ratio disagreements on Klaeger involve profiles with 14–27 active kinases, and six of these have one clearly dominant target (top_1–top_2 gap above 1). The ratio ranks these seven 31st to 63rd of 200 and the entropy and Gini 146th to 180th: the ratio reflects the dominant target, and the entropy and Gini the number of active kinases. Bosc et al.[8] described such profiles for individual compounds. On Davis, the most unstable compound overall is a profile of this kind (Figure 2, right).

Monotonicity Under Weak-Target Addition

Table 4 summarizes the criteria. None of the four definitions has an eligibility rule, and any score can adopt one. No boundary parameter meets D2 on every dataset, and the verdicts are the same for any ε up to 13% (Supporting Information, Table S3). The published entropy

and $P_{\max}$ have no boundary parameter and are not assessed against D2. D3 separates the scores.

Table 4: The three criteria for the four definitions, three comparators and the gated entropy. $P_{\max}$ and the published entropy have no boundary parameter and are assessed in the text. ✓: satisfied. ×: violated. D1 is assessed from each score's reporting rule, D2 empirically on the four datasets (Methods), and D3 from the functional form. The comparators are reported only for compounds with an active target.

	S-score		Count	Entropy		Gini	Ratio	Gated
	$\tau > x_0$	$\tau = x_0$		$\beta = x_0$	$\beta = \tau^*$			entropy
D1 (eligibility)	×	✓	✓	×	✓	×	×	✓
D2 (consistency)	×	×	×	×	×	×	×	×
D3 (monotonicity)	×	✓	✓	×	✓	×	✓	✓

The S-score violates D3 for any threshold above the floor: adding a kinase bound between the floor and the threshold increases N but not the number of hits, so the compound looks more selective. With the threshold at the floor, every addition that D3 allows is a hit, and a compound with C hits among N kinases moves from $-C/N$ to $-(C+1)/(N+1)$, which is not higher. The active count ranks every panel as the S-score at τ^* does but does not divide by N, so it satisfies D3. D3 therefore concerns how scores change when a panel is extended, not how compounds are ranked on one panel. The Gini violates D3, since a weakly bound kinase can raise its inequality. The ratio satisfies it, because a weak addition cannot displace the primary target or lower the reference off-target. $P_{\max}$ satisfies it too: a weak addition leaves its numerator unchanged and raises its denominator. The published entropy violates it: with floor-coded pairs given no weight, a profile with one target at $\mathrm{p}K_d = 6.01$ and fifty at 5.01 scores -5.35 bits, and adding one at 6.0 raises this to -5.19.

The entropy satisfies D3 at $\beta = \tau^*$, where weak binding carries no weight, but not at $\beta < \tau^*$. Its hard cutoff gives a kinase just above β a weight close to zero, so a profile with many such kinases has high entropy, and adding one kinase bound more strongly, but still below τ^*, concentrates the distribution. With $\beta = 5.0$, a profile with one kinase at $\mathrm{p}K_d = 7.0$ and fifty at 5.1 scores -4.89 bits, and adding one at 6.0 raises this to -4.83. Adding one

kinase at τ^* never raises the entropy's score on Davis, Klaeger or Metz at any baseline in their sweeps. On Anastassiadis it does at baselines below the 20% floor (33% of the compounds at $\beta = 0\%$, 16% at 5%). Adding real kinases bound below the threshold to random 50- or 100-kinase panels raises the score in at most 0.1% of cases on any dataset.

The gated entropy satisfies D1 by construction and D3 by the proposition in Methods, and its weights vary smoothly with affinity above the gate. D3 also holds for every scored compound on all four datasets when one kinase is added at τ^*. Its score falls with the number of active kinases (Spearman -0.78 to -0.97); without the non-binder gate the floor-coded pairs dominate and the score rises with it ($r = +0.91$ on Klaeger). It ranks the compounds almost exactly as the entropy does (0.994–0.998) and as the S-score with $\tau = x_0$ does (0.998–0.9997), which also satisfies D1 and D3. Moving τ^* from 5.5 to 7 (40% to 70% on Anastassiadis) changes which compounds are scored but not how those scored at both values are ranked (Spearman ≥ 0.999). Under the noise model of Methods, D1 changed the status of at most 11 compounds in any draw, and the ranking of the compounds scored with and without noise stayed at Spearman 0.978 or higher with the noise-free ranking, against 0.983 or higher for the entropy. This tests noise in measured values, not uncertainty about which floor-coded pairs bind above the floor. The guarantee concerns adding a target: strengthening a target already in the profile can raise the score (one target at 7 and fifty at 5.1, with one of the fifty raised to 6.0, moves from -5.642 to -5.637). What the gated entropy adds is a Shannon entropy with a proof of D3 and weights that vary smoothly above the gate; on these data it is not an empirical improvement on simpler scores. Variants with other Rényi orders, smoothing widths and floors are in Supporting Information (Section S4).

Recovery from Small Panels

On small random panels, some compounds can no longer be scored. At 50 targets, 88% of the compounds scored on the full panel are still scored on Davis, 71% on Klaeger, 77% on Anastassiadis and 99% on Metz (Figure 3, top). The compounds lost are mostly those

with few active targets: on a 50-target Klaeger subpanel, a compound with one active target stays scored with probability 0.23, and one with ten with probability 0.93. High conditional agreement on a small panel can therefore coexist with the loss of many narrowly active compounds. Coverage limits the recovery of pair order. Keeping 90% of clearly separated pairs takes 150 targets on Davis, 200 on Klaeger and Anastassiadis, and 40 on Metz for every score except the ratio, the published entropy and $P_{\max}$, which need as many or more (Figure 3, middle).

Among compounds scored on both panels, scores whose ranking depends mostly on the few strongest targets recover slowest (Figure 3, bottom). The S-score with $\tau = x_0$, the entropy with $\beta = x_0$, the Gini and the gated entropy rank compounds almost by the share of targets with detected binding (Spearman above 0.97 with the S-score at $\tau = x_0$ on every dataset). They reach p^* at 20 targets on Davis, 50 on Klaeger and 30 on Anastassiadis. The S-score at τ^* and the active count need 30, 80 and 50, and the entropy with $\beta = \tau^*$ 40, 100 and 65. On Metz, where untested targets are set to the floor, the S-score at τ^* and the active count reach p^* at 50 targets, before the share-following scores (65). With bars of 0.80, 0.85 or 0.95 instead of 0.90, the share-following scores still reach the bar no later than the S-score and the entropy at τ^* on Davis, Klaeger and Anastassiadis, and the ratio, the published entropy and $P_{\max}$ are the slowest scores on every dataset. The published entropy and $P_{\max}$, with floor-coded pairs given no weight, need 300 of 363 targets on Davis and 200 of 217 on Klaeger; on Metz the published entropy needs 150 of 172 and $P_{\max}$ more than the largest subpanel tested (150). The ratio needs 200 of 217 targets on Klaeger and more than the largest subpanel tested on the other datasets (300 of 363, 250 of 291 and 150 of 172). The published entropy gives weak binding a small positive weight, so including weak binding does not by itself make a score recover quickly. The gated entropy recovers rankings no better than the S-score with $\tau = x_0$.

p^* is a threshold on the mean: at p^*, 62–90% of individual subpanels exceed 0.90 for the share-following scores (Supporting Information, Section S5). The active count and the

S-score at τ^* give identical recovery on every dataset, although only the count satisfies D3. On Davis, Anastassiadis and Metz, compounds whose score changes most when one target is left out tend to have larger rank errors at 50 targets (Spearman $r = 0.21$–0.73 for the entropy with $\beta = x_0$, the Gini and the gated entropy); on Klaeger this association is weak ($r \leq 0.14$).

Discussion

The scores compared here summarize different aspects of a binding profile, and how far their rankings agree depends on choices made before any score is computed. The S-score, entropy and Gini agree more with one another than with the ratio when the entropy and Gini are computed from readouts above a common baseline and compounds without an active kinase are left out. With association-constant weights, or with compounds that have no active kinase included, this grouping weakens or disappears. With most entries imputed at the floor, as on Metz, the three scores agree because they follow how many kinases were measured. Ratio measures match the pairwise definition of selectivity in analytical chemistry, a ratio of binding constants for an analyte and an interferent.[22] Bosc et al.'s window score also correlates strongly with the ratio, and their ranking score is a ratio by definition (Supporting Information, Section S6). The ratio and the S-score, entropy and Gini disagree most on broad profiles with one dominant target. A selectivity ranking is therefore a statement about a score, its implementation, and a set of compounds and targets, not about the compounds alone. On compounds and kinases shared by two datasets, changing the definition moved rankings about as much as changing the dataset.

Reporting Recommendations

Each recommendation follows from a result above.

1. State the target universe and the score implementation, including the weighting and any baseline.

2. Distinguish measured affinities, pairs with no detected binding, rejected fits and untested pairs, and state the value and weight each receives.

3. Apply D1: state the activity threshold, withhold or flag the scores of compounds with no target above it, and rank the eligible compounds among themselves. Otherwise an active compound's instability depends on how the ineligible compounds are scored, as the Gini–skewness association showed.

4. Fix the activity boundary in the analysis protocol and report how rankings change when it is moved. A compound's rank standard deviation across the sweep summarizes this, but it is not a guide to robustness across datasets: on the 46 shared compounds it is not associated with how far a compound's rank moves between Davis and Klaeger ($|r| \leq 0.18$; Supporting Information, Section S6).

5. For small panels, report how many compounds remain scorable as well as how well the ranking of those compounds is recovered.

6. When comparing studies that use different scores, fix the comparison protocol in advance and report the compounds on which the scores disagree.

Limitations

The four datasets cover three assay technologies but a small part of chemical and kinase space, and they detect different shares of active interactions. Missing values are not random: Metz compounds were tested against different kinases. On no dataset is the floor a measured detection limit. It codes non-detection on Davis; non-detection, low-confidence fits and weak measured affinities on Klaeger; and untested pairs on Metz. On Anastassiadis it is an operational cutoff for weak inhibition. Fixing the boundary at the floor therefore depends on how each dataset codes these pairs. Random subpanels are one sampling design; panels chosen by kinase family, as many screening panels are, may recover rankings differently. Cell-free panels can also miss interactions that cellular target-engagement assays detect.[23,24]

No score considered here uses target identity or exposure. Two compounds with the same affinity distribution receive the same score whether their off-targets are inert or linked to a class toxicity, and an affinity alone does not establish whether a target is occupied at a tolerated dose. A selectivity score is at best a summary of the binding profile, to be weighed with target-specific liability knowledge and exposure modeling. Selectivity rankings of the approved Klaeger compounds were not associated with adverse-event report proportions or label discontinuation rates (Supporting Information, Section S7).

The criteria do not single out one score, and on these data no score was preferable on every criterion. The gated entropy's guarantee covers only the addition of a weakly bound target. The criteria are also unlikely to be complete: a measure should be correctly oriented against a crude proxy such as the number of active kinases, since a measure can satisfy D1 and D3 and still rank promiscuous compounds as selective.

Conclusion

Kinase inhibitor selectivity rankings depend on choices usually left implicit: how affinities are weighted, which compounds are ranked, how pairs without detected binding are treated, which targets the panel includes, and where the activity boundary is placed. These choices change whether the common scores agree, how much a ranking moves when the activity boundary moves, and how much of a ranking a small panel retains. Selectivity analyses should state them, apply an activity-eligibility rule, fix the activity boundary in advance and report its effect, and report how many compounds remain scorable on small panels. The gated entropy shows that a Shannon entropy whose weights vary smoothly above a fixed gate can be made monotone under weak-target addition, but on these data it ranks compounds almost as the share of targets with detected binding does.

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Data and Software Availability

All data and software used in this study are publicly available. The Davis kinase inhibitor profiling dataset is available at `https://github.com/dingyan20/Davis-Dataset-for-DTA-Prediction`. The Klaeger et al. dataset is available as supplementary data to the original publication (Science, 2017, `https://doi.org/10.1126/science.aan4368`). The Anastassiadis et al. (*Nat. Biotechnol.* 2011) and Metz et al. (*Nat. Chem. Biol.* 2011) datasets are available as supplementary data to their original publications (`https://doi.org/10.1038/nbt.2017`; `https://doi.org/10.1038/nchembio.530`). All analysis scripts, processed data files, and the LaTeX source for this manuscript are available at `https://github.com/polinavino/kinase-selectivity-definitions`; this repository includes the code and full per-definition results for the secondary clinical-outcome analysis (FAERS adverse-event report rates, obtained via openFDA queries, and FDA-label discontinuation rates, from FDA product labels, for the FDA-approved Klaeger compounds). No confidential data were used in this study. All results reported in this manuscript are fully reproducible from the pub-

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Conflict of Interest

The author declares no conflict of interest.

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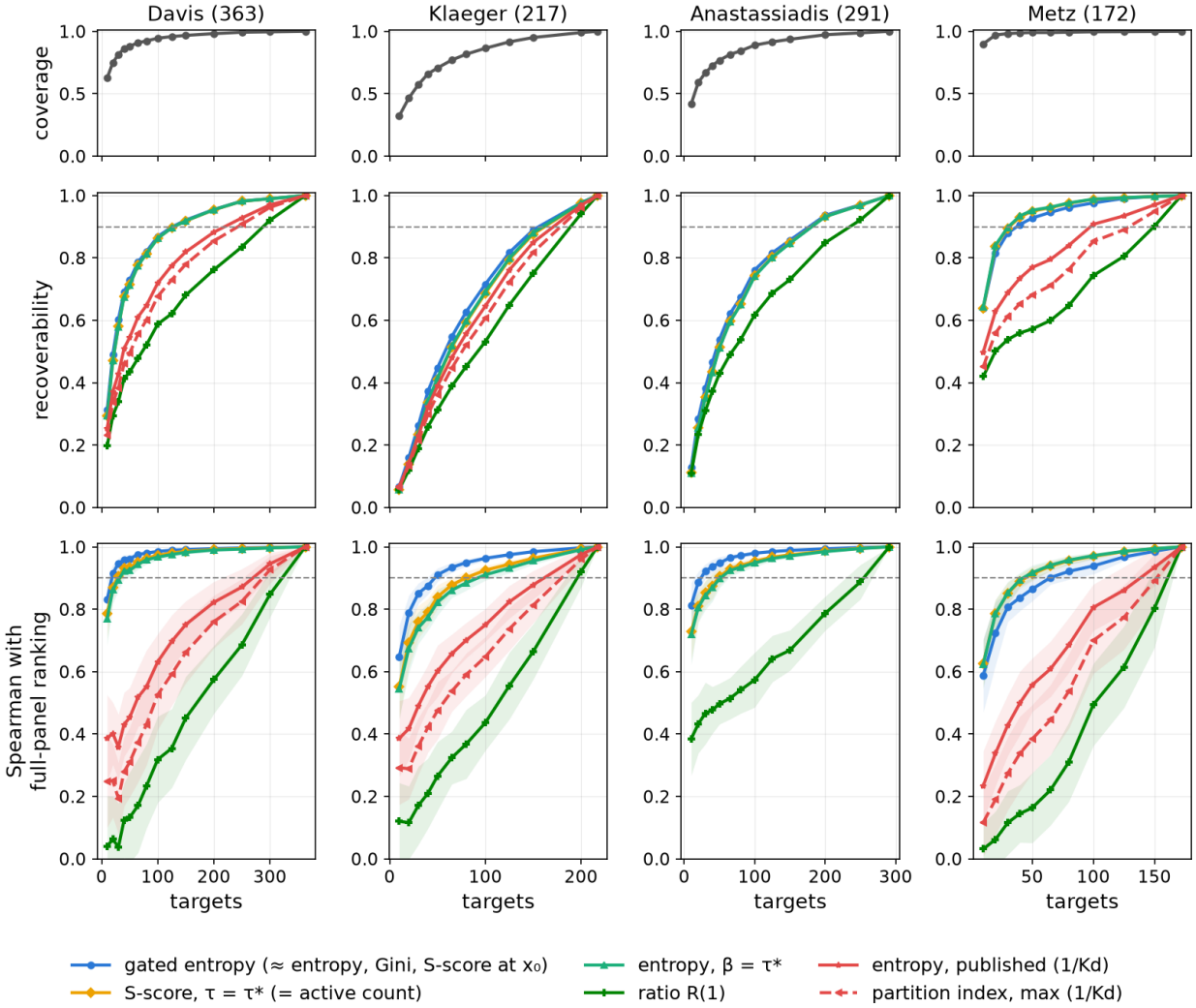


Figure 3: Recovery from random subpanels with D1 applied to every score, on all four datasets (50 subpanels per size). Top: coverage, the share of compounds scored on the full panel that are still scored. Middle: recoverability, the share of clearly separated full-panel pairs that are scored on the subpanel and keep their order. Bottom: mean Spearman correlation between subpanel and full-panel rankings over compounds scored on both, with ± 1 standard deviation across subpanels shaded. Each score is compared with its own full-panel ranking. The ratio is $R(1)$. The published entropy and $P_{\max}$ give floor-coded pairs no weight and are not defined on Anastassiadis. The S-score at τ^* and the active count coincide. The entropy and Gini with $\beta = x_0$ and the S-score with $\tau = x_0$ are not drawn: their curves differ from the gated entropy's by at most 0.015. Dashed line: 0.90.